

CHAPTER : 2

Introduction:

The entities required for the Myntra Database Management System are identified based on the business requirements of the organization. The attributes of each entity, along with their primary keys, foreign keys and data types are defined to ensure proper data organization and efficient data management. The relationships between the entities are identified to represent customer management, product management, seller management, inventory management, order processing, payment management and delivery management. In addition, the cardinality of each relationship is determined to define how the entities are associated with one another. This analysis provides a clear understanding of the database requirements and establishes a strong foundation for designing an efficient relational database while ensuring data integrity, consistency and minimal redundancy.

2.0 Customer:

Attribute	Data Type	Key	Description
Customer_ID	INT	PK	Unique customer ID
First_Name	VARCHAR(50)	-	Customer first name
Last_Name	VARCHAR(50)	-	Customer last name
Gender	VARCHAR(10)	-	Gender
Email	VARCHAR(100)	-	Email
Phone	VARCHAR(15)	-	Phone
Password	VARCHAR(100)	-	Login password
Address	TEXT	-	Address
City	VARCHAR(50)	-	City
State	VARCHAR(50)	-	State
Pincode	INT	-	Postal code
Registration_Date	DATE	-	Registration date

2.1 Product:

Attribute	Data Type	Key	Description
Product_ID	INT	PK	Unique product ID
Product_Name	VARCHAR(100)	-	Product name
Category_ID	INT	FK	References Category
Seller_ID	INT	FK	References Seller
Brand	VARCHAR(50)	-	Brand
Size	VARCHAR(20)	-	Available size
Color	VARCHAR(30)	-	Product color
Price	DECIMAL(10,2)	-	Product price
Discount	DECIMAL(5,2)	-	Discount
Stock	INT	-	Available stock

2.2 Category:

Attribute	Data Type	Key	Description
Category_ID	INT	PK	Unique category ID
Category_Name	VARCHAR(50)	-	Category name
Category_Description	TEXT	-	Category description
Status	VARCHAR(20)	-	Active / Inactive

2.3 Seller:

Attribute	Data Type	Key	Description
Seller_ID	INT	PK	Unique seller ID
Seller_Name	VARCHAR(100)	-	Seller name
Contact_Person	VARCHAR(50)	-	Contact person
Phone	VARCHAR(15)	-	Phone
Email	VARCHAR(100)	-	Email
Address	TEXT	-	Address
GST_Number	VARCHAR(20)	-	GST number

2.4 Inventory:

Attribute	Data Type	Key	Description
Inventory_ID	INT	PK	Unique inventory ID
Product_ID	INT	FK	References Product
Quantity	INT	-	Available quantity
Warehouse	VARCHAR(100)	-	Warehouse
Last_Updated	DATE	-	Last updated

2.5 Orders:

Attribute	Data Type	Key	Description
Order_ID	INT	PK	Unique order ID
Customer_ID	INT	FK	References Customer
Order_Date	DATE	-	Order date
Total_Amount	DECIMAL(10,2)	-	Total amount
Order_Status	VARCHAR(30)	-	Placed/Shipped/Delivered
Shipping_Address	TEXT	-	Shipping address
Delivery_Date	DATE	-	Delivery date

2.6 Order Items:

Attribute	Data Type	Key	Description
Order_Item_ID	INT	PK	Unique order item ID
Order_ID	INT	FK	References Orders
Product_ID	INT	FK	References Product
Quantity	INT	-	Quantity
Unit_Price	DECIMAL(10,2)	-	Unit price
Discount	DECIMAL(5,2)	-	Discount
Total_Price	DECIMAL(10,2)	-	Total price

2.7 Warranty:

Attribute	Data Type	Key	Description
Warranty_ID	INT	PK	Unique warranty item ID
Product_ID	INT	FK	References product
Warranty_Period	VARCHAR(50)	-	Warranty duration
Warranty_Start	DATE	-	Warranty start date
Warranty_End	DATE	-	Warranty expiry date
Warranty Status	VARCHAR(30)	-	Active/Expired

2.8Payment:

Attribute	Data Type	Key	Description
Payment_ID	INT	PK	Unique payment ID
Order_ID	INT	FK	References Orders
Payment_Method	VARCHAR(30)	-	UPI/Card/COD
Transaction_ID	VARCHAR(100)	-	Transaction ID
Payment_Status	VARCHAR(20)	-	Status
Payment_Date	DATE	-	Payment date
Amount	DECIMAL(10,2)	-	Amount

2.9Review:

Attribute	Data Type	Key	Description
Review_ID	INT	PK	Unique review ID
Customer_ID	INT	FK	References customers
Product_ID	INT	FK	References

Rating	INT	-	Rating(1-5)
Review Text	TEXT	-	Customer review
Review Date	DATE	-	Date of review

3.0 Shipment:

Attribute	Data Type	Key	Description
Shipment_ID	INT	PK	Unique shipment ID
Order_ID	INT	FK	References Orders
Courier_Name	VARCHAR(100)	-	Shipping partner
Tracking_Number	VARCHAR(100)	-	Tracking number
Shipment_Status	VARCHAR(30)	-	Packed / Shipped / Out for Delivery / Delivered / Returned
Dispatch_Date	DATE	-	Date shipped
Delivery_Date	DATE	-	Date delivered

3.1Admin:

Attribute	Data Type	Key	Description
Admin_ID	INT	PK	Unique admin ID
Admin_Name	VARCHAR(100)	-	Admin name
Email	VARCHAR(100)	-	Email
Phone	VARCHAR(15)	-	Phone
Role	VARCHAR(30)	-	Role
Username	VARCHAR(50)	-	Username
Password	VARCHAR(100)	-	Password

3.2 Relationships:

Entity 1	Relationship	Entity 2	Cardinality	Explanation
Customer	Places	Orders	1 : M	One customer can place many orders, but each order belongs to one customer.
Orders	Contains	Order Items	1 : M	One order contains multiple products.
Product	Appears In	Order Items	1 : M	A product can appear in many order items.
Category	Contains	Product	1 : M	One category contains many products.
Seller	Sells	Product	1 : M	One seller sells many products.
Product	Stored In	Inventory	1 : 1	Each product has one inventory record.
Orders	Has	Payment	1 : 1	Every order has one payment record.
Orders	Delivered By	Delivery	1 : 1	Each order is delivered through one delivery record.
Admin	Manages	Product	1 : M	Admin manages and updates multiple products.
Admin	Manages	Orders	1 : M	Admin manages and monitors multiple orders.

Conclusion:

The **Myntra Database Management System** is designed to efficiently manage customer, product, seller, inventory, order, payment, and shipment information. It ensures data accuracy, reduces redundancy, and supports smooth e-commerce operations. This database provides a reliable and scalable solution for managing Myntra's online shopping system.